4

Kendall & Kendall
Systems Analysis and Design, 9e

Information Gathering: Interactive Methods

Kendall & Kendall Copyright © 2014 Pearson Education, Inc. Publishing as Prentice Hall

Objectives

- Recognize the value of interactive methods for information gathering.
- Construct interview questions to elicit human information requirements and structure them in a way that is meaningful to users.
- Understand the purpose of stories and why they are useful in systems analysis.
- Understand the concept of JAD and when to use it.
- Write effective questions to survey users about their work.
- Design and administer effective questionnaires.

Kendall & Kendall Copyright © 2014 Pearson Education, Inc. Publishing as Prentice Hall

Interactive Methods to Elicit Human Information Requirements

- Interviewing
- Joint Application Design (JAD)
- Questionnaires

Kendall & Kendall Copyright © 2014 Pearson Education, Inc. Publishing as Prentice Hall

Major Topics

- Interviewing
  - Interview preparation
  - Question types
  - Arranging questions
  - The interview report
- User Stories
- Joint Application Design (JAD)
  - Involvement
  - Location
- Questionnaires
  - Writing questions
  - Using scales
  - Design
  - Administering

Kendall & Kendall Copyright © 2014 Pearson Education, Inc. Publishing as Prentice Hall

Interviewing

- Interviewing is an important method for collecting data on human and system information requirements
- Interviews reveal information about:
  - Interviewee opinions
  - Interviewee feelings
  - Goals

Kendall & Kendall Copyright © 2014 Pearson Education, Inc. Publishing as Prentice Hall

Interview Preparation

- Reading background material
- Establishing interview objectives
- Deciding whom to interview
- Preparing the interviewee
- Deciding on question types and structure

Kendall & Kendall Copyright © 2014 Pearson Education, Inc. Publishing as Prentice Hall

Question Types

- Open-ended
- Closed

Kendall & Kendall Copyright © 2014 Pearson Education, Inc. Publishing as Prentice Hall

Open-Ended Questions

- Open-ended interview questions allow interviewees to respond how they wish, and to what length they wish
- Open-ended interview questions are appropriate when the analyst is interested in breadth and depth of reply

Kendall & Kendall Copyright © 2014 Pearson Education, Inc. Publishing as Prentice Hall

Advantages of Open-Ended Questions

- Puts the interviewee at ease
- Allows the interviewer to pick up on the interviewee's vocabulary
- Provides richness of detail
- Reveals avenues of further questioning that may have gone untapped

Kendall & Kendall Copyright © 2014 Pearson Education, Inc. Publishing as Prentice Hall

Advantages of Open-Ended Questions (continued)

- Provides more interest for the interviewee
- Allows more spontaneity
- Makes phrasing easier for the interviewer
- Useful if the interviewer is unprepared

Kendall & Kendall Copyright © 2014 Pearson Education, Inc. Publishing as Prentice Hall

Disadvantages of Open-Ended Questions

- May result in too much irrelevant detail
- Possibly losing control of the interview
- May take too much time for the amount of useful information gained
- Potentially seeming that the interviewer is unprepared
- Possibly giving the impression that the interviewer is on a “fishing expedition”

Kendall & Kendall Copyright © 2014 Pearson Education, Inc. Publishing as Prentice Hall

Closed Interview Questions

- Closed interview questions limit the number of possible responses
- Closed interview questions are appropriate for generating precise, reliable data that is easy to analyze
- The methodology is efficient, and it requires little skill for interviewers to administer

Kendall & Kendall Copyright © 2014 Pearson Education, Inc. Publishing as Prentice Hall

Benefits of Closed Interview Questions

- Saving interview time
- Easily comparing interviews
- Getting to the point
- Keeping control of the interview
- Covering a large area quickly
- Getting to relevant data

Kendall & Kendall Copyright © 2014 Pearson Education, Inc. Publishing as Prentice Hall

Disadvantages of Closed Interview Questions

- Boring for the interviewee
- Failure to obtain rich detailing
- Missing main ideas
- Failing to build rapport between interviewer and interviewee

Kendall & Kendall Copyright © 2014 Pearson Education, Inc. Publishing as Prentice Hall

Attributes of Open-Ended and Closed Questions (Figure 4.5)

Kendall & Kendall Copyright © 2014 Pearson Education, Inc. Publishing as Prentice Hall

Bipolar Questions

- Bipolar questions are those that may be answered with a “yes” or “no” or “agree” or “disagree”
- Bipolar questions should be used sparingly
- A special kind of closed question

Kendall & Kendall Copyright © 2014 Pearson Education, Inc. Publishing as Prentice Hall

Probes

- Probing questions elicit more detail about previous questions
- The purpose of probing questions is:
  - To get more meaning
  - To clarify
  - To draw out and expand on the interviewee's point
- May be either open-ended or closed

Kendall & Kendall Copyright © 2014 Pearson Education, Inc. Publishing as Prentice Hall

Arranging Questions

- Pyramid
  - Starting with closed questions and working toward open-ended questions
- Funnel
  - Starting with open-ended questions and working toward closed questions
- Diamond
  - Starting with closed, moving toward open-ended, and ending with closed questions

Kendall & Kendall Copyright © 2014 Pearson Education, Inc. Publishing as Prentice Hall

Pyramid Structure

- Begins with closed questions
- Expands by allowing open-ended questions and more generalized responses
- Is useful if interviewees need to be warmed up to the topic or seem reluctant to address the topic

Kendall & Kendall Copyright © 2014 Pearson Education, Inc. Publishing as Prentice Hall

Pyramid Structure for Interviewing Goes from Specific to General Questions (Figure 4.7)

Funnel Structure

- Begins with generalized, open-ended questions
- Concludes by narrowing the possible responses using closed questions
- Provides an easy, nonthreatening way to begin an interview
- Is useful when the interviewee feels emotionally about the topic

Kendall & Kendall Copyright © 2014 Pearson Education, Inc. Publishing as Prentice Hall

Funnel Structure for Interviewing Begins with Broad Questions then Funnels to Specific Questions (Figure 4.8)

Diamond Structure

- A diamond-shaped structure begins in a very specific way
- Then more general issues are examined
- Concludes with specific questions
- Combines the strength of both the pyramid and funnel structures
- Takes longer than the other structures

Kendall & Kendall Copyright © 2014 Pearson Education, Inc. Publishing as Prentice Hall

Diamond-Shaped Structure for Interviewing Combines the Pyramid and Funnel Structures (Figure 4.9)

Closing the Interview

- Always ask “Is there anything else that you would like to add?”
- Summarize and provide feedback on your impressions
- Ask whom you should talk with next
- Set up any future appointments
- Thank them for their time and shake hands.

Kendall & Kendall Copyright © 2014 Pearson Education, Inc. Publishing as Prentice Hall

Interview Report

- Write as soon as possible after the interview
- Provide an initial summary, then more detail
- Review the report with the respondent

Kendall & Kendall Copyright © 2014 Pearson Education, Inc. Publishing as Prentice Hall

Joint Application Design (JAD)

- Joint Application Design (JAD) can replace a series of interviews with the user community
- JAD is a technique that allows the analyst to accomplish requirements analysis and design the user interface with the users in a group setting

Kendall & Kendall Copyright © 2014 Pearson Education, Inc. Publishing as Prentice Hall

Conditions That Support the Use of JAD

- Users are restless and want something new
- The organizational culture supports joint problem-solving behaviors
- Analysts forecast an increase in the number of ideas using JAD
- Personnel may be absent from their jobs for the length of time required

Kendall & Kendall Copyright © 2014 Pearson Education, Inc. Publishing as Prentice Hall

Who Is Involved

- Executive sponsor
- IS analyst
- Users
- Session leader
- Observers
- Scribe

Kendall & Kendall Copyright © 2014 Pearson Education, Inc. Publishing as Prentice Hall

Benefits of JAD

- Time is saved, compared with traditional interviewing
- Rapid development of systems
- Improved user ownership of the system
- Creative idea production is improved

Kendall & Kendall Copyright © 2014 Pearson Education, Inc. Publishing as Prentice Hall

Drawbacks of Using JAD

- JAD requires a large block of time to be available for all session participants
- If preparation or the follow-up report is incomplete, the session may not be successful
- The organizational skills and culture may not be conducive to a JAD session

Kendall & Kendall Copyright © 2014 Pearson Education, Inc. Publishing as Prentice Hall